

Levels-of-Processing Effects on a Variety of Memory Tasks: New Findings and Theoretical Implications

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Three experiments investigated level of processing (LOP) effects on a variety of direct and indirect memory tasks, in the context of a processing theory of dissociations. Subjects studied words in five encoding conditions and received one of ten memory tests. In Experiment 1, four tests previously classified as conceptual showed a robust LOP effect, as did a direct perceptual test of graphemic cued recall. An indirect perceptual word fragment completion test was unaffected by LOP. Experiment 2 showed that a new indirect version of a graphemic cued test was not affected by LOP. In Experiment 3, guided by a generation/recognition model, we constructed three new direct tests in which subjects identified words that were graphemically, phonologically, or semantically similar to studied words. The three tests differed in their sensitivity to study conditions, but LOP had no effect in any case, despite the involvement of deliberate conscious recollection. Contemporary explanatory frameworks couched as dichotomies (e.g., implicit/explicit, perceptual/conceptual) do not provide an adequate account of the results. It seems necessary instead to specify the types of information activated by each encoding condition, the types of information required by each test, and how encoding and retrieval processes are modified by task instructions. © 1996 Academic Press, Inc.

INTRODUCTION

Two principles that account for a large number of effects in memory research are, first, the notion that qualitatively different types of encoding are differentially effective for later memory performance and, second, the notion that given a specific type of encoding, retrieval is optimized when the retrieval test is designed to utilize that same type of information. The first principle was embodied in the levels-of-processing (LOP) framework presented by Craik and Lockhart (1972), and the second principle of a necessary compatibility between encoded information and information required by a test is central to the ideas of encoding specificity (Tulving & Thomson, 1973) and transfer appropriate processing (Morris, Bransford, & Franks, 1977). Early

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work on levels of processing established that memory is better for material studied under semantic than physical (or perceptual) encoding conditions (e.g., Craik & Tulving, 1975). The initial studies involved recall and recognition, the popular tests of the day. The finding has been replicated many times and is viewed as the standard LOP effect. The effect provided important empirical support for the original LOP framework and the idea that the processing of the same material can be “deeper” or “shallower” with deeper processing leading also to better performance in a subsequent memory test.

In further work it was shown that study and test conditions can be arranged so that the standard LOP effect disappears or is even reversed. The early demonstrations were accomplished by using explicit tests² designed to benefit from nonsemantic encoding, for example rhyme-cued recall or rhyme recognition test (e.g., Fisher & Craik, 1977; Morris et al., 1977). It should be noted, however, that these studies also showed that a rhyme encoding coupled with a rhyme retrieval task is associated with poorer performance than the combination of semantic encoding and semantic retrieval.

More recently, LOP has played an important role in the study of implicit memory. Initially, researchers reported that LOP does not affect implicit (or “indirect”) memory tests, in contrast to the robust effect on conventional explicit or “direct” tests (e.g., Graf & Mandler, 1984; Jacoby & Dallas, 1981). This conclusion has been modified, however, as researchers developed a more sophisticated understanding of implicit memory: One useful distinction is that between perceptual and conceptual tests (e.g., Blaxton, 1989; Jacoby, 1983; Roediger, Weldon, & Challis, 1989; Tulving & Schacter, 1990). Perceptual tests are those in which subjects try to resolve perceptually impoverished or degraded stimuli, as with word fragment completion (-l-ph--t), or word identification (e.g., “elephant” is flashed on a screen very briefly). In contrast, conceptual tests provide an intact test cue that bears a semantic or conceptual relation to the studied event and subjects are asked to provide an appropriate response, as with category instance generation (animal ____), or answering general knowledge questions (What animal did Hannibal use to cross the Alps?). The categorization of indirect tests into perceptual and conceptual revealed a reasonably consistent pattern of effects. First, it was noted that early reports of no effects of LOP on implicit memory involved perceptual tests (later work, however, showed that small effects can be found—see Brown & Mitchell, 1994; Challis & Brodbeck, 1992). Second, fewer LOP studies have involved indirect conceptual tests (e.g., Challis & Sidhu, 1993; Hamann, 1990; Srinivas & Roediger, 1990), but these have consistently shown a robust LOP effect similar to the effect in conventional direct tests of recall and recognition.

The general pattern of LOP effects on direct and indirect tests can be explained by a processing theory of performance on memory tests (e.g., Blaxton, 1989; Roe-

² Throughout this paper we use the terms “explicit” and “implicit” tests (or tasks) to refer to retrieval conditions that involve either deliberate, conscious recollection of the original episode (explicit) or conditions in which the original event affects later test performance in the absence of any conscious intention to retrieve (implicit). The terms “direct” and “indirect” are also used to describe these respective tasks (Richardson-Klavehn & Bjork, 1989). In all cases, we acknowledge that a given retrieval task can involve both conscious and unconscious influences, in the sense of Jacoby, Toth, and Yonelinas (1993).

diger, 1990; Roediger et al., 1989). A basic tenet of the theory is that performance on a test reflects the similarity of the mental operations promoted at study and required by the test, an idea embodied in the principle of transfer appropriate processing (Morris et al., 1977). With this in mind, the theory explains LOP effects on indirect tests in the following way. In the typical LOP experiment, subjects see a word and perform a semantic task (e.g., make a meaning-based judgment) or a physical task (e.g., search for a specific letter). The semantic task promotes meaning-based processing, compared to the physical task. So, those tests that depend on conceptual information for their completion (i.e., indirect conceptual tests; conventional direct tests) benefit more from semantic than physical study manipulations. In a similar vein, reports of no (or small) LOP effects on indirect perceptual tests are consistent with the idea that these tests depend on perceptual information and the standard LOP manipulation does not affect the encoding of such information.

Despite this general understanding of the effects of LOP on implicit and explicit memory performance, the present research base is severely limited with respect to both encoding and retrieval conditions. Many researchers have used only one shallow encoding condition (e.g., a graphemic or rhyme judgment) and one deep semantic condition (e.g., pleasantness rating or living/nonliving judgment), whereas many other interesting encoding manipulations exist (e.g., the encoding of self-referent information). Similarly, the choice of retrieval tests has often been restricted to recall and recognition in the domain of explicit memory and to perceptual identification and word fragment and word stem completion in the implicit domain. The processing theory asserts that the classification of perceptual or conceptual is independent of the distinction between direct and indirect memory tests; rather, direct and indirect tests can be perceptual or conceptual in nature. One reason for carrying out the first experiment in the present series was simply to broaden the empirical base by using a wider selection of encoding tasks in combination with a more representative selection of direct and indirect retrieval tasks.

What pattern of results might be expected from such a combination of encoding and retrieval conditions? Clearly, a processing perspective predicts that test performance will reflect the similarity of mental operations promoted at study and required by the retrieval task, but further predictions depend on how the underlying representations are organized. If, for example, qualitatively different types of information are processed by a mosaic of parallel, domain-specific modules (see Fodor, 1983), we might expect good performance when encoding and retrieval operations are compatible, but poor performance otherwise. If four qualitatively different types of encoding are designated a,b,c,d, and their corresponding compatible retrieval tests are designated A,B,C, and D, respectively, then performance should be good when encoding a is combined with retrieval A (also in cases bB, cC, dD) but poor in other cases (e.g., aC, bA, cD). This simple model is pattern 1A in Table 1.

Table 1 also sketches various other possibilities. If compatible encoding/retrieval combinations are *not* in fact equivalent, but a depth of processing dimension exists such that deeper encoding conditions (in the order a,b,c,d, say) confer an advantage provided that their potential is tapped by a compatible retrieval task (see Moscovitch & Craik, 1976), then the pattern labeled 1B should be found—that is, compatible combinations yield good performance, but deeper combinations are better than

TABLE 1
Possible Patterns of Memory Performance When Four Levels of Encoding
Are Combined with Four Retrieval Tests

Pattern	Retrieval tests	Encoding Tasks			
		a	b	c	d
1A	A	1	0	0	0
	B	0	1	0	0
	C	0	0	1	0
	D	0	0	0	1
1B	A	1	0	0	0
	B	0	2	0	0
	C	0	0	3	0
	D	0	0	0	4
1C	A	1	1	1	1
	B	0	2	2	2
	C	0	0	3	3
	D	0	0	0	4
1D	A	1	2	3	4
	B	2	4	6	8
	C	3	6	9	12
	D	4	8	12	16

Note. The symbols a,b,c,d represent four different encoding tasks varying from shallow (a) to deep (d) encoding. The symbols A,B,C,D represent four retrieval tests compatible with a,b,c, and d, respectively. The values in the table represent comparative levels of memory performance. See text for fuller details.

shallow combinations. When these LOP ideas are brought into play and added to the compatibility principle, one possibility is that each retrieval test requires a certain level of encoded information but cannot utilize deeper information. Another option is that each successively deeper level of encoding incorporates earlier (or shallower) levels. Thus, test B requires encoding b, but does not do any better with levels c and d. This scheme yields pattern 1C. The final possibility is that the progressively more favorable encoding conditions a,b,c,d benefit *all* retrieval tasks, but that deeper tests (e.g., C,D) benefit to a greater degree (pattern 1D).

Some of the theoretical schemes shown in Table 1 are more realistic than others. For example, we know that 1A is implausible from the findings of Fisher and Craik (1977) as well as Morris et al. (1977); that is, combination dD gives better performance than combination aA. But 1B is also unlikely as a general model given the graded LOP effects typically found in recall and recognition (Craik & Tulving, 1975). Pattern 1C may be a plausible model for implicit perceptual tests in light of the negligible LOP effects found in such tests. Finally, pattern 1D is also plausible—especially for explicit retrieval tasks—in the sense that it allows for a graded interaction of compatibility and LOP effects. In any event, Table 1 suggests various possible

patterns of memory performance; the patterns found empirically will shed further light on the combination of LOP and compatibility between encoding and retrieval. The results will also address questions of cognitive organization; for example whether mental activities are organized as operations within parallel modules or are also organized on the vertical dimension of mental functioning, i.e., whether they are essentially hierarchically organized processes. A strong version of this last hypothesis implies that deeper levels of processing are in control of operations at earlier, or shallow, levels.

With these considerations in mind we designed Experiment 1, in which subjects studied words under five different encoding conditions and then retrieved the words by means of one of six memory tests. The encoding conditions involved orienting tasks designed to induce various levels of encoding; we also included an intentional learning condition. The retrieval tests included both direct and indirect tests and also tests requiring perceptual or conceptual information. We thus hoped to address theoretical questions relevant to a processing account of memory (e.g., Craik, 1983; Roediger, 1990; Roediger et al., 1989) and to recent views on the hierarchical organization of memory and cognition (e.g., Lockhart & Craik, 1990; Velichkovsky, 1994).

EXPERIMENT 1

Experiment 1 was a large-scale LOP experiment in which subjects studied lists of words in five study conditions and then received one of six memory tests. Four incidental study tasks were designed to promote the encoding of different types of information: (a) letter-search promoted processing of visual graphemic information; (b) counting of syllables promoted phonological processing; (c) making a living/nonliving decision promoted semantic processing; and (d) judging the relevance of the word to oneself promoted encoding of self-referent information. One intentional study condition was included for comparison purposes. There was also a nonstudied baseline condition.

The six memory tests represented different types of test. Five of the tests were the same as those used by Blaxton (1989). These included indirect tests of word fragment completion and answering general knowledge questions, and direct tests of graphemic cued recall, semantic cued recall, and free recall. The sixth test was yes/no recognition. These tests cover the four cells of the conventional indirect/direct by conceptual/perceptual matrix: word fragment completion is an indirect perceptual test, general knowledge is an indirect conceptual test, graphemic cued recall is a direct perceptual test, and semantic cued recall, free recall, and recognition are direct conceptual tests.

An important reason for using Blaxton's tests is that she provided empirical support for the classification of these tests. She showed that a number of study variables affected the perceptual tests in the same way, but differently from the conceptual tests. So, for instance, modality of presentation had an effect on word fragment completion and graphemic cued recall but did not affect the conceptual tests (general knowledge, semantic cued recall, free recall). The perceptual tests were sensitive to manipulations of the physical form of the stimuli at study and test, whereas the conceptual tests were sensitive to semantic encoding. Blaxton (1989) did not examine

the effects of LOP on the collection of tests. Given the popularity of this manipulation it is important to see if LOP can be used for operationally defining whether a memory test is perceptual or conceptual in nature, as suggested by some (e.g., Hamann, 1990). That is, does LOP affect conceptual tests but not perceptual tests, regardless of whether they are direct or indirect tests? There has been some research that speaks to the issue (e.g., Thapar & Greene, 1994, Hamann, 1990) but the relevant work has been limited in scope (as discussed earlier).

Experiment 1 addressed the issue in a systematic fashion by using a variety of study conditions and administering a battery of tests previously classified as being conceptual or perceptual and direct (explicit) or indirect (implicit). We expected to find evidence of compatibility between encoding and retrieval tasks and also confirmation of the LOP manipulation. With regard to the latter, however, we expected different effects of LOP on tests requiring perceptual as opposed to conceptual information and speculated that LOP might show different patterns on direct and indirect tests.

Method

Subjects. The participants were 108 students enrolled in an undergraduate psychology course at the University of Toronto. They participated as part of a course requirement.

Design and materials. The experiment was a mixed factorial, with study condition as a within-subject factor and test type as a between-subject factor. The five study conditions were letter-count, syllable-count, living/nonliving, self-referent, and intentional. There was also a nonstudied baseline condition. The tests were word fragment completion, general knowledge, semantic cued recall, graphemic cued recall, free recall, and recognition. Eighteen subjects were assigned to each of the six test conditions.

A set of 108 target words and their corresponding test items were selected from the materials used by Blaxton (1989). For each target (e.g., copper), there was a corresponding word fragment (c-pp--), a general knowledge question (What makes up 10% of yellow gold?), a semantically related word (bronze), and a graphemically similar word (chopper). The 108 targets were randomly separated into six lists of 18 items. Two buffer words were included at the beginning and at the end of each list. The six lists of 22 words were typed in lowercase letters on separate sheets of paper. For each subject, one list was assigned to each of the five study conditions and one list was assigned to the nonstudied condition. Across subjects, the six lists were counterbalanced across the study and nonstudied conditions, and the order of study conditions was counterbalanced.

The tests of word fragment completion, general knowledge, semantic cued recall, and graphemic cued recall were constructed by typing the 108 cues on test sheets. For the recognition tests, the 108 target words were randomly mixed with 216 distractor words. Each target had 1 semantically related and 1 graphemically related word associated with it; the resulting 324 words were typed on test sheets. Test items were typed in lowercase letters and presented in a random order. In free recall, responses were written on sheets of lined paper.

Procedure. Subjects were tested as a group during a regular class, except for seven subjects who were tested individually. At the onset of the experimental session, sub-

jects were given a booklet of materials. They were told to read the instructions carefully and listen to the instructor for additional instructions. The cover sheet informed subjects that they would perform a number of different tasks involving words. Subjects proceeded through the booklet as directed by the instructor. They read some general instructions and then proceeded through the booklet. For each study condition, they were given 2 min to read the instructions and 2 min to perform the task. In the case of the letter-count condition, subjects counted the number of ascending and descending letters in each word (i.e., letters such as “t” and “y” that extend above or below the main body of the word). In the syllable-count condition, they counted the number of syllables in each word. In the living/nonliving judgment condition, they indicated whether a word represents a living thing or a nonliving object. In the self-referent condition, they were instructed to “think about the importance of each word to you now or in the future.” They made an importance rating on a 5-point scale, with 1 corresponding to no importance and 5 to very important. In the above study conditions, subjects wrote their response beside each word; they were not informed that memory would be tested for these words. In the intentional condition, subjects were told to study the list of words for a later memory test. They were encouraged to do their best to learn the words. They were told to stop studying the words after the allotted time. The study phase lasted about 25 min.

After the study phase, the test booklets were distributed to subjects. The cover sheet on each booklet provided the test instructions. The instructions were modeled after those used by other researchers (e.g., Blaxton, 1989; Challis & Sidhu, 1993). For the word fragment completion test, subjects were instructed to complete each word fragment so that it made an English word. For the general knowledge test, they were instructed to answer each question with one word. On fragment completion and general knowledge, no mention was made of the study phase. On the other tests, subjects were explicitly told that they were receiving a memory test for all of the words presented earlier. It was emphasized that this was a memory test for all of the words presented as part of the different study tasks. In the case of semantic and graphemic cued recall, subjects were told that they would be provided with cues to help them remember words presented earlier. For semantic cued recall, subjects were instructed to think about the meaning of each cue word and to use that to help them remember a word presented earlier. For graphemic cued recall, they were instructed to look at the overall shape or pattern formed by the letters of each cue and to use that to help them remember a word. They were told that the overall shape or form of the letters was the important thing and the meaning of the word was irrelevant and should be ignored. In the aforementioned tests, responses were written on the test sheets beside each test cue. Subjects were paced through the test by giving them 12 s per item. On the recognition test, subjects were told that words were presented on the attached sheets, some of which were presented earlier and others were not. They were instructed to think carefully about each word and circle those words presented earlier. In the case of free recall, subjects were instructed to recall the words presented earlier. They were encouraged to recall as many words as possible and if they were unsure whether a word was presented earlier, they should write it down. Free recall and recognition were self-paced; subjects were given 22 min to complete these tests. An experimental session lasted about 1 hr.

TABLE 2
Proportions of Target Words Produced or Recognized as a Function
of Encoding Condition and Test Type (Experiment 1)

Test type	LSD	Encoding Condition					
		Nonstudied	Letter	Syllable	Living	Self	Intentional
Indirect							
Word fragment completion	.08	.22	.39	.44	.43	.41	.41
General knowledge test	.08	.19	.22	.25	.31	.41	.38
Direct							
Graphemic cued recall	.07	.05	.06	.16	.23	.31	.23
Semantic cued recall	.06	.05	.08	.15	.21	.29	.23
Free recall	.06	.00	.04	.07	.17	.24	.33
Recognition	.11	.12	.27	.51	.70	.83	.72

Results and Discussion

The results are presented in Table 2, which displays the proportion of target words produced or recognized as a function of study condition and test type. The four incidental orienting tasks are shown in order of increasing depth on the basis of previous studies; letter, syllable, living, and self-referent. It is unclear where the intentional learning task should come in this order, so it is shown displaced to the right. The retrieval tasks are shown in Indirect and Direct groups; within these groups they are ordered from perceptual to conceptual. For each test, a one-way analysis of variance (ANOVA) that included the five study conditions and the nonstudied baserate condition was performed on the proportion of targets produced or recognized. The degrees of freedom for these analyses were 5 and 85. For all tests, the analyses yielded a significant effect so differences between conditions were evaluated using a least significance difference test (LSD). The level of significance for all tests reported in this article was set at .05.

In word fragment completion, the proportion of fragments completed in the five study conditions was greater than in the nonstudied condition (i.e., priming occurred), and performance did not differ among the five study conditions ($F = 9.63$, $MSe = .013$, $LSD = .08$). In the general knowledge test, there was significant priming in the living/nonliving, intentional, and self-referent conditions. Performance in the self-referent condition was reliably greater than the semantic and intentional condition; these two conditions were not reliably different ($F = 9.42$, $MSe = .015$, $LSD = .08$). In semantic cued recall, performance was best in the self-referent condition, followed by the intentional and living/nonliving conditions, then the syllable condition, and last the letter condition ($F = 22.71$, $MSe = .007$, $LSD = .06$). Graphemic cued recall showed the same pattern of performance as semantic cued recall ($F = 17.87$, $MSe = .011$, $LSD = .07$). Free recall improved across the letter, syllable, living/nonliving, self-referent, and intentional conditions. The letter and syllable conditions were not reliably different ($F = 39.39$, $MSe = .007$, $LSD = .06$). In recognition, the proportion of target words correctly identified was reliably greater in the self-referent condition than the other conditions. The intentional and living/nonliving

TABLE 3

Differences between Encoding Conditions for Each Test Type in Experiment 1 (see Table 2 and text)

	Encoding condition				
	Letter	Syllable	Living	Intentional	Self
Indirect					
Word fragment completion	2	2	2	2	2
General knowledge test	0	0	1	2	2
Direct					
Graphemic cued recall	0	1	2	2	3
Semantic cued recall	0	1	2	3	4
Free recall	0	1	2	5	4
Recognition	1	3	5	5	6

Note. Each number in the table represents the difference between that encoding condition and the nonstudied baseline in LSD steps for each test type.

conditions, which were not reliably different, were better than the syllable condition; these were better than the letter condition. In all conditions, hit rate exceeded false alarms ($F = 40.27$, $MSe = .032$, $LSD = .12$). The recognition test included distractor items that were graphemically similar or semantically related to the target words. The proportions of these items incorrectly identified as studied words varied from .03 to .08 across the various condition; there were no reliable differences ($F_s < 1.93$).

The results can be presented in another way that highlights the effects of LOP on the various tests. Table 3 shows the data from Table 2, but presented as the number of LSDs above the nonstudied baseline for each retrieval test. Thus, for graphemic cued recall, the Letter condition was not reliably different from baseline. Syllable was more than 1 LSD above baseline. Living exceeded 2 LSDs above baseline and so on. Table 3 shows that the variation of encoding conditions had an effect on every test except word fragment completion. Further, the LOP effect is clearly a graded one in all cases, with the possible exception of General Knowledge, where the effect may asymptote at 2 LSDs above baseline. One speculative possibility is that indirect retrieval tests require a certain level of encoding; shallower levels may provide the required type to some extent, but the test cannot utilize deeper or more elaborately encoded information than its “target” level. Thus, word fragment completion requires graphemic information which is equally present in all five encoding conditions, whereas General Knowledge requires semantic information which is not present in Letter and Syllable encoding but is present in the remaining three conditions, most adequately in Self-referent and Intentional conditions.

With respect to the remaining direct tests, they are all typically regarded as requiring conceptual information for optimal performance, so one interpretation of Table 3 is that the four incidental encoding conditions provide semantic information which increases from Letter to Self encoding. The notion that encoded semantic information can be more or less elaborate (e.g., Craik & Tulving, 1975) is well accepted, but it is not at all clear why syllable encoding should involve more semantic activity than counting letters or why judgments of personal relevance should recruit more semantic associations than semantic categorization itself. In any event, other investigations

show that phonological, semantic, and metacognitive modes of processing are qualitatively different (e.g., Levelt, 1989; Shimamura, 1994; Velichkovsky, 1990, 1995). An unqualified attribution of LOP effects to semantic processing alone would be dubious also in light of other aspects of our data. For instance, free recall and recognition benefit from study conditions in rather different ways. Whereas it might be argued that free recall utilizes conceptual information to a greater degree, it is recognition that shows the steeper increase in performance as a function of the LOP manipulation. In addition, recognition deviates from baseline performance even after letter encoding, although this study condition involves little if any semantic processing.

Two further points may be noted from Table 3. The first is that Intentional learning gave rise to a level of performance between Living and Self in most cases. This result is in line with many previous findings showing that a semantic incidental orienting task can give as good memory performance as intentional learning (Craig & Lockhart, 1972) and that Self-referent encoding is a particularly effective form of encoding (e.g., Miall, 1986; Rogers, Kuiper, & Kirker, 1977). The one exception is Free Recall, for which intentional learning is best; this result may be attributed to the likelihood that intentional instructions invoke organizational strategies of particular benefit to this form of retrieval (Brown, 1975). The second point is that whereas the two indirect tests yield patterns somewhat in line with pattern 1C in Table 1, the direct tests more clearly fit pattern 1D. We return to these points later.

The pattern of LOP effects fits with Blaxton's (1989) results and classification of the tests, with one exception. First consider the compatible findings: The conceptual tests (direct and indirect) were affected by LOP, whereas the indirect perceptual test was not affected by LOP. In line with Blaxton's findings, a study manipulation of a semantic nature affected conceptual tests but not a perceptual test. The one inconsistent finding was that graphemic cued recall, a supposedly perceptual test, showed a graded LOP effect much like the conceptual tests. The differential effect of LOP on word fragment completion and graphemic cued recall is at odds with Blaxton's (1989) work, which provided rather compelling support for the processing view that word fragment completion and graphemic cued recall are perceptual tests, and that perceptual tests, whether direct or indirect, are affected in a similar way. In light of the dissociative effect of LOP on these two tests (see also Thapar & Greene, 1994), we must question these assumptions and try to understand the supposed discrepancy, which we do in Experiment 2.

EXPERIMENT 2

Previous results show that graphemic cued recall behaves much like word fragment cued recall. On the one hand, both tests are affected by physical manipulations (e.g., modality of presentation) consistent with a perceptual test (e.g., Blaxton, 1989; Challis, Chiu, Kerr, Law, Schneider, Yonelinas, & Tulving, 1993; Craig, Moscovitch, & McDowd, 1994). On the other hand, both tests are strongly affected by LOP, consistent with a conceptual test (our Experiment 1, for graphemic cued recall; Roediger, Weldon, Stadler, & Riegler, 1992, for word fragment cued recall). Furthermore, researchers have shown that LOP affects word fragment cued recall and word stem

cued recall but not word fragment completion (e.g., Challis & Tulving, 1993; Craik et al., 1994; Roediger et al., 1992). That is, an indirect word fragment cued test consistently behaves as a perceptual test, whereas a direct word fragment test behaves like a perceptual or conceptual test, depending on the study manipulation. This finding implies that the nature of the test instructions (direct or indirect) can affect the type of information that contributes to performance on a test. This state of affairs can be explained in the context of processing theory.

Performance on word fragment cued tests (e.g., fragment completion, cued recall) can be explained quite well by a generation/recognition model, which has been described in some detail (Challis & Tulving, 1993; Jacoby & Hollingshead, 1990; Roediger et al., 1992). The basic idea is that the presentation of word fragment cues invokes retrieval akin to generation, and generation is driven by the perceptual nature of the cue so that the probability of retrieving a studied target is sensitive to the physical similarity of the stimulus form at study and test. Generation underlies performance in word fragment completion in that subjects can respond appropriately with any solution that comes to mind. In word fragment cued recall, subjects are instructed to complete the fragment with a studied word, so there must be generation followed by a second, recognition process in which studied items are selected from those generated. The recognition process is sensitive to semantic encoding and to other postperceptual forms of processing. In this way word fragment cued recall is sensitive to study manipulations like LOP.

The similarities between word fragment cued recall and graphemic cued recall, in relation to word fragment completion, imply that performance on graphemic cued recall can be interpreted in terms of the generation/recognition model. If this is the case, we would expect that when we present graphemic test cues with indirect retrieval instructions ("graphemic cued generation"), performance should not be affected by LOP (like word fragment completion in Experiment 1). Experiment 2 examined the effects of LOP on graphemic cued generation.

Method

Subjects. Twenty-four students enrolled in an undergraduate psychology course at the University of Toronto participated as part of a course requirement.

Design, materials, and procedure. The study conditions, materials, and procedure were the same as in Experiment 1. The test materials were the same as those used for graphemic cued recall in Experiment 1, except that three blank lines were placed beside each test word. The test instructions told subjects to look at the overall shape or pattern formed by the letters of a word and write down three words with a similar shape of form. They were told that the meaning of the words was irrelevant and should be ignored. As one example, they were told they may look at "eagle" and come up with words like "maple," "eager," or "signal." They were told not to be critical about the words they wrote down, for there was no right or wrong response. Subjects were given 10 s with each word.

Results and Discussion

The proportions of target words produced in the five study conditions were .18 (letter-search), .21 (syllable-count), .21 (living/nonliving), .22 (self-referent), and .18

(intentional). The nonstudied baseline condition was .11. A one-way ANOVA that included the study and nonstudied conditions yielded a significant effect, $F(5, 115) = 5.24$, $MSe = .01$. Further analyses ($LSD = .06$) indicated that the proportions of targets produced in the five study conditions were not reliably different and that all exceeded the nonstudied baseline value (i.e., there was significant priming in all five encoding conditions.)

Priming in graphemic cued generation was not affected by the LOP manipulation, in sharp contrast to the robust LOP effect on graphemic cued recall in Experiment 1. With precisely the same graphemic test cues, LOP affected performance on a direct but not an indirect test. Similarly, LOP has a robust effect on word fragment cued recall in Experiment 1 but little, if any, effect on word fragment completion (Challis & Tulving, 1993; Craik et al., 1994; Roediger et al., 1992).

The pattern of findings can be interpreted in terms of a generation/recognition model of performance on perceptually cued tests. With graphemic cues (as word fragment cues), the indirect retrieval process (generation) is guided by the perceptual nature of the cue. So, priming from the prior study of a target word is affected by manipulations of the perceptual similarity of the stimulus form at study and test, but not by postperceptual manipulations of the kind promoted by our LOP variables. If we administered direct cued recall instructions with these same perceptual cues (e.g., graphemic cued recall, word fragment cued recall), the same indirect retrieval process (generation) occurs, along with a recognition process by which studied items are selected from those generated on the basis of all available evidence. The recognition process is sensitive to encoded phonology, meaning, and metacognitive attitudes, so that graphemic cued recall is sensitive to LOP in a systematic fashion. The graphemic cued recall test was originally constructed by Blaxton (1989) as an example of an explicit data-driven test; she showed that the test behaved like word fragment completion, an implicit perceptual test. Blaxton's study variables were physical manipulations so that graphemic cued recall behaved like a perceptual test. From Experiment 1 we know that with a broader manipulation of postperceptual study variables, graphemic cued recall behaves like a conceptual test.

The fact that graphemic cued recall is affected by LOP calls into question Blaxton's (1989) assertion that graphemic cued recall is a direct data-driven test (at least in the same way that word fragment completion is a data-driven test). The finding also raises the question as to whether one can construct a direct perceptual test that is not sensitive to LOP. The point is an important one, for one of the basic tenets of the processing view is that data-driven (i.e., perceptual) tests are not limited to indirect tests but can also include direct tests. We considered that the construction of a retrieval task involving direct instructions (i.e., deliberate, conscious recollection of previous events), but that was not sensitive to LOP, would reveal further insights into the compatibility of encoding and retrieval processes. With this in mind, Experiment 3 was motivated by our quest to design a direct (explicit) test that would not be sensitive to LOP.

EXPERIMENT 3

Our main insight into constructing a direct test that would not be sensitive to LOP came from the generation/recognition model of performance on perceptually cued

tests. The gist of our reasoning was this: The test must be designed so that retrieval of the studied word depends on the overlap in perceptual information in the test cue and the memory trace, which means that the test cue and studied target word must bear a physical resemblance. Second, the test must be designed to obviate the use of a strategy whereby subjects rely on recognition to select studied words from possible responses (because recognition of studied words would produce an LOP effect). In tests like graphemic cued recall, subjects are told to respond only with studied words, so prior to responding, one obvious strategy is to rely on recognition to verify that an item was actually studied. These considerations suggested that we design a direct test (a) in which the test cue and target word were physically similar and (b) that did not require subjects to respond with a studied target word.

Accordingly, the following test was designed. Subjects were presented with test words that were graphemically similar to studied target words (e.g., people and purple). The target and test words were those used in the graphemic cued recall test in Experiment 1. During the recognition test, subjects were instructed to identify those words that had a similar physical shape or form to a studied word. The test is called a “graphemically similar recognition test.” We felt the test could be performed in two ways. On the one hand, the test cue may directly retrieve a physically similar word encoded in memory, with successful retrieval depending only on the physical information available in the test cue and memory trace. If this was the case, we would not expect an LOP effect, on the assumption that similar perceptual information is encoded in the various study conditions. On the other hand, if subjects use test cues to generate physically similar words and then use recognition to select a studied words from those generated (like graphemic cued recall), then we would expect an LOP effect that is attributable to the recognition process.

Having constructed the graphemically similar recognition test, we constructed two additional tests for the sake of comparison purposes. In working with the graphemic items, we noticed that target words and their corresponding test cue tended to be phonologically similar (e.g., midget and fidget). Given this similarity, we constructed a “phonologically similar recognition test.” The materials were the same as those used in the graphemically similar test; subjects were told to identify those words that sounded like a studied word. The third test was recognition of semantically similar words. The test contained words that were semantically related to studied target words (e.g., study = bronze; test = copper); these were the words used in the semantic cued recall test in Experiment 1. Subjects were told to identify those words whose meaning was similar or related to a studied word. The test list in all three cases was identical; half of the words were related graphemically to studied words and half were related semantically to studied words. It was expected, of course, that the “graphemically similar” instruction would lead to recognition of words of the first type.

In sum, subjects in Experiment 3 studied words in the five encoding conditions used in the previous experiments and were then administered one of three direct memory tests, a graphemically similar, phonologically similar, or semantically similar recognition test.³ With respect to the theoretical patterns shown in Table 1, we

³ It should be noted that similar direct retrieval tests in which test words bear some specified resemblance to previously studied words have also been used (although for different purposes) by Jacoby (1975), Morris et al. (1977), and by Marmurek (1995) among others.

predicted that the different encoding conditions would affect these three direct tests in different ways. Consider recognition of graphemically similar words. On the assumption that visual (graphemic) information underlies performance, we expected similar performance across the study conditions because seeing the target word in all study conditions invokes encoding of the relevant graphemic information. The LOP effect would therefore be similar to that obtained for indirect word fragment completion (Experiment 1) and graphemic cued generation (Experiment 2). In contrast, we expected that recognition of phonologically similar words would be poor following the letter-counting task because phonological information is unlikely to be encoded, but that performance would be higher (and equivalent) following the other encoding tasks. By the same logic, performance on the semantically similar task was predicted to be good following any study task involving semantic encoding, but relatively poor in other cases. That is, the overall pattern of results might approximate pattern 1C in Table 1.

Method

Subjects. The participants were 72 students enrolled in an undergraduate psychology course at the University of Toronto. They participated as part of a course requirement.

Design, materials, and procedure. The five study conditions were the same as those in Experiments 1 and 2. Type of test was varied between subjects, with three groups of 24 subjects, each completing one of the three recognition tests (graphemically similar, phonologically similar, or semantically similar). The three groups were given identical materials at test (words related graphemically or semantically to studied words); the groups differed only in terms of test instructions, as described below.

From the materials used in Experiment 1, we selected the 108 target words and their corresponding graphemically similar and a semantically related words (e.g., freckle–fickle–birthmark). The study materials and procedure were the same as in Experiments 1 and 2. For the test, the 108 graphemically similar words and the 108 semantically related words were printed in lowercase, and in a random order, on four test sheets. The same four test sheets were given to subjects in all three groups. The cover sheet of the test booklet provided the instructions. Subjects were told they were receiving a memory test for the words presented earlier. They were informed that the attached sheets contained many words and that none of the words were presented earlier but that half of the words were similar in some respect to the studied words. For the graphemically similar recognition test, subjects were told that half of the words looked like studied words and they should focus on the overall shape of the word and ignore its meaning. For the phonological recognition test, subjects were told that half of the words sounded like studied words and they should focus on the sound and ignore the meaning. For the semantic test, subjects were told that half of the words had a similar meaning to a studied word and that they should focus on the meaning and ignore the physical form of the word. For the three test groups, subjects were instructed to go through the words and to select 70 words that met the criterion, and to circle the best 70 words. Subjects were instructed to select a set number of words so as to maintain a similar response criterion across the three test

TABLE 4
Recognition of Words Graphemically (A), Phonologically (B), and Semantically (C)
Similar to Studied Words as a Function of Encoding Condition and Item Type

Item type	Encoding condition					
	Nonstudied	Letter	Syllable	Living	Intentional	Self
A. Graphemic test						
Graphemically similar	.27	.37	.41	.39	.40	.41
Semantically related	.23	.25	.28	.25	.23	.26
B. Phonological test						
Graphemically similar	.29	.29	.39	.45	.45	.41
Semantically related	.25	.27	.27	.25	.24	.29
C. Semantic test						
Graphemically similar	.26	.25	.24	.22	.26	.26
Semantically related	.35	.32	.35	.45	.46	.45

Note. Values significantly above the nonstudied baseline are printed in bold.

groups. A pilot study showed that subjects were capable of selecting 70 words in the allotted time. They were told they had 15 min to perform the task, although they were given 20 min, at the end of which all subjects indicated that they had completed the task.

Results and Discussion

Table 4 shows the proportions of target words identified as a function of test type, study condition, and item type. For each test type, the top row corresponds to those words that were graphemically similar to target words, and the bottom row corresponds to those words semantically related to target words. For each recognition test, separate one-way ANOVAs (that included the studied and nonstudied conditions) were performed on the graphemically similar and semantically related items. If the analysis yielded a significant effect, a LSD was computed to assess differences among the studied and nonstudied conditions.

For the graphemic test, the one-way analysis was significant for the graphemically similar items, $F(1, 115) = 4.58$, $MSe = .014$. The LSD of .07 indicated that the five study conditions were not reliably different but that all studied conditions exceeded the nonstudied baseline condition, thereby establishing memory for the studied words. A one-way ANOVA on the semantically related items was not significant, $F < 1$. For the phonological test, a one-way ANOVA was significant for the graphemically similar items, $F(5, 115) = 9.63$, $MSe = .014$. The LSD of .07 indicated that the letter and nonstudied conditions were not reliably different, and significantly lower than the other four study conditions (syllable, living, self, and intentional); these four conditions did not differ among themselves. A one-way ANOVA on the semantically related items was not significant, $F < 1$. For the semantic test, a one-way ANOVA was significant for semantically related items, $F(1, 115) = 5.41$, $MSe = .015$. A LSD of .07 indicated that the living, self, and intentional conditions were significantly greater than the nonstudied, letter, and syllable conditions; there was no difference

among the three conditions in the two groupings. A one-way ANOVA on the graphemically related items was not significant, $F < 1$.

Several main findings are clearly apparent in Table 4: First, subjects were, as instructed, able to identify words that were graphemically, phonologically, or semantically similar to studied words. Second, the results showed a systematic LOP effect, in the following sense: If we view the five study conditions as a sequence (letter-search, syllable-count, living/nonliving categorization, intentional, and self-relevant), then there was a critical encoding condition after which all conditions showed similar memory performance for the studied words (e.g., syllable counting for the phonological test and living/nonliving categorization for the semantic test). Comparing across the three tests, the range of study conditions in which words were reliably identified was affected by LOP, but the level of performance within any one study condition was not affected by LOP. The flat performance across these study conditions is atypical and contrasts sharply with the graded LOP effect obtained on one indirect and several direct tests in Experiment 1. Returning to one of the issues raised in the discussion of Experiment 1, the point at which study conditions begin to affect performance seems to be of importance.

One purpose of Experiment 3 was to construct a direct test that is not sensitive to LOP, like word fragment completion in Experiment 1. The results met our expectations in that recognition of graphemically similar words was not affected by the LOP manipulation. This finding is consistent with the view that recognition of graphemically similar words was based on encoded perceptual information. Seeing the target word in any one of the five study conditions apparently involved similar perceptual analyses, independent of the encoding task, so that performance was not affected by the LOP manipulation. This same assumption has often been made in explaining the lack of a LOP effect on indirect perceptual tests (Roediger et al., 1992; Tulving & Schacter, 1990).⁴

On the phonological recognition test, words were identified from all study conditions except the letter-search condition. In the graphemic test, however, words were equally well identified from all five study conditions. The difference between the phonological and graphemic tests implies that the two tests measured different kinds of processing. In the context of the present discussion, the dissociation indicates that studying a word in the letter-search task induced the encoding of graphemic but not phonological information, so that subjects failed to identify words studied in the letter-search condition. Furthermore, recognition of phonologically similar words in the other four study conditions implies that phonological information was encoded in these conditions.

Turning to the semantic recognition test, subjects were able to identify semantically related words from three study conditions (semantic, intentional, and self-relevant). This finding fits with the idea that memory performance in this test depends on seman-

⁴ Clearly this analysis is based on a rather undifferentiated notion of perceptual processing. One can easily find perceptual encoding which, like spatial localization (cf. Bridgeman, Van der Heijden, & Velichkovsky, 1994; Velichkovsky, 1982), will be worse in terms of later memory performance than the identity-based encodings used in LOP studies so far. Such a contrast of different perceptual encodings should lead to typical LOP effects even in perceptual memory tests.

tic information and that such information was encoded in the three study conditions. The fact that words from the letter-search and syllable-count conditions were not identified implies that little if any semantic information was encoded in these conditions. This conclusion supports our previous analysis of the results of Experiment 1.

Overall, the results of Experiment 3 show that subjects can recognize words that are graphemically, phonologically, or semantically related to previously presented target words, but only (a) if the target words are encoded in a way that involves the relevant processing dimension and (b) if subjects are instructed to treat the cue words in the appropriate (i.e., dimension- or domain-specific) fashion. When these conditions are met, recognition performance is significantly above baseline, but the LOP manipulation has no further effect. Thus, at first sight, Experiment 3 seems to demonstrate a pure compatibility effect between the type of information encoded and that required for successful retrieval. A more detailed inspection, however, shows that the pattern of data combines features of strongly parallel (1A in Table 1) and weakly hierarchical (1C in Table 1) models. In the General Discussion we contrast the results from the three experiments with a view to understanding when different forms of LOP effects do and do not appear.

GENERAL DISCUSSION

In the three experiments, subjects studied words under five different encoding conditions; memory for these studied words was subsequently tested on a total of 10 different retrieval tasks. The resulting combinations yielded a number of data patterns which, while complex, nonetheless fall into three broad categories. One is the "traditional" LOP pattern in which successive encoding conditions are associated with progressively higher levels of performance. This graded LOP effect approximates the pattern "1 2 3 4 5" in the terminology of Table 1 and was shown most clearly by the direct retrieval tasks in Experiment 1. The second pattern is one in which all encoding conditions exceeded the baseline but the LOP manipulation has no further effect within the studied range of encoding. This pattern ("1 1 1 1 1" or "2 2 2 2 2") was shown by word fragment completion in Experiment 1 and by indirect graphemic cued recall in Experiment 2. The third pattern is novel and was detected in the present study only because of the large-scale combination of encoding and retrieval conditions. In this pattern, shallow processing conditions lead to memory performance that does not exceed baseline; at some point a condition does give performance that reliably exceeds the baseline, but subsequent encoding conditions do not show further improvement. That is, rather than being graded or flat, the pattern is stepped, for example taking the form "01111" or "00111" as in the phonologically similar and semantically similar recognition tests in Experiment 3.

One important point made by these results is that experimental designs using only two encoding conditions are likely to miss the true underlying patterns. For example, if Experiment 3 had been conducted with only the letter-search and living/nonliving encoding conditions, we would have concluded that the graphemically similar test showed no LOP effects, whereas the phonological and semantic tests did show such effects. This conclusion would have failed to note the stepped functions illustrated clearly in Table 4.

A related point—and this is one we wish to stress—is that contemporary explanatory frameworks couched as dichotomies fail to give an adequate account of the present results. As one example, the reported experiments were designed largely in the context of a processing account of memory performance (Blaxton, 1989; Roediger, 1990; Roediger & McDermott, 1993; Roediger et al., 1989). This theoretical approach makes a distinction between perceptual and conceptual tests, and applies the idea of transfer appropriate processing, leading to the prediction that perceptual tests will benefit from a study condition that promotes perceptually based processing, whereas conceptual tests will benefit more from a study condition that promotes meaning-based processing. It is assumed that the distinction between perceptual and conceptual tests supersedes the distinction between direct and indirect tests; that is, direct and indirect perceptual tests will behave similarly to each other, but differently from direct and indirect conceptual tests, which do resemble each other. Several findings from the present study contradict this account. First, in Experiment 1, graphemic cued recall (a perceptual test) showed a graded LOP effect that was quite similar to the effect shown by conceptual tests. Second, in this same experiment, graphemic cued recall behaved very differently from word fragment completion, despite Blaxton's (1989) claim that these two tests behave in a similar fashion. Third, when graphemic cued recall was given with indirect retrieval instructions in Experiment 2, it now *did* behave like word fragment completion. Current versions of processing theory would have some difficulty accounting for a test that behaves like a perceptual test under one set of instructions, but like a conceptual test under other instructions.

The direct/indirect distinction does not fare any better. A number of studies in the literature suggest that retrieval tasks involving deliberate recollection are sensitive to the LOP manipulation, whereas indirect tasks are not (Craig et al., 1994; Graf & Mandler, 1984; Jacoby & Dallas, 1981). It is certainly the case that the direct tasks used in Experiment 1 do show the graded pattern, whereas the indirect tasks in Experiment 1 and 2 do not (with the debatable exception of the general knowledge test). The problem for this account comes in Experiment 3; in this case *all* the recognition tests were intentional or direct, yet none of the three crucial conditions showed traditional graded LOP effects.

However, any account that focuses on the levels-of-processing approach alone is also clearly invalidated by the present results. If it is agreed that the five encoding conditions represent different points on some shallow–deep continuum, then a simple depth of processing model should predict a similar pattern of effects across different retrieval tasks. Experiments 1, 2, and 3 make it clear that this does not happen. Instead, it seems that any satisfactory account must consider (a) the encoding task and the type or types of information about the stimulus that it invokes and therefore encodes; (b) the retrieval task, its informational requirements, and its compatibility with the encoding task; and (c) the instructions given to subjects at both encoding and retrieval. With regard to this last point, it does not seem that intentionality plays any special role at either encoding or retrieval; rather, intentional learning or intention to retrieve a previously presented item simply leads to the subject switching on processes that he or she believes to be optimal for the task. Such processes may also be activated “unintentionally” by the task requirements themselves. Thus in Experiment 1, Intentional encoding leads to performance that lies between Living and Self

encoding in five of six cases; the exception is Free Recall in which, as already noted, intentionality is likely to be associated with organizational processing that benefits free recall. Similarly, at retrieval, the processes activated are simply those that subjects believe will be effective for the task as described (e.g., generation of any word or retrieval of a list item). Such processes may also be activated “unintentionally” by the task requirements themselves (see Boucart, Lorenceau, & Humphreys, 1995, for an analysis of unintentional activation of semantic representations in processing of perceptual dimensions of an object).

In place of such dichotomous accounts of memory performance, we propose the following set of ideas. First, along with most current researchers, we assume that neither encoding tasks nor retrieval tasks are “process pure” in the sense that they involve only one type of domain-specific processing. Rather, each task, while targeting one primary or dominant “code,” may well involve other types to some degree. Thus, a syllable encoding task will emphasize phonological processing, but will also involve other codes (e.g., graphemic in our case). Similarly, a recognition retrieval test may draw primarily on encoded semantic information, but will also benefit significantly from graphemic and phonological encodings. In these terms, the theoretical problem becomes one of specifying the subset of underlying processes (or types of information) activated by each encoding condition, the subset of processes that each retrieval task draws on, and how the subset can be changed by instructions. We also suggest that the data patterns associated with the various encoding/retrieval combinations can reveal the functional architecture of the underlying processing. As an example, if deeper levels of encoding involve the activation of specific shallower codes (as shown by their effects on retrieval tasks sensitive to these shallower codes), it could be concluded that the various processing codes are not strictly parallel, but are organized in a hierarchical fashion, with lower perceptual operations feeding into higher cognitive processes.

In line with this analysis and following Bernstein’s (1947) work on levels of motor control, it is useful to differentiate the leading level from background coordination in any task. In Bernstein’s terms, the leading level is defined by the intentional goal of the task, whereas background coordination consists of a context of supportive operations, usually at lower levels of control. For example, in our syllable encoding condition the leading level is that of phonological processing, but the task also demands some degree of background visual processing. If a retrieval task, like recognition, can draw on information at both levels—visual perceptual and phonological—then retrieval task performance will benefit more from syllable-counting (phonological plus visual processing) than from letter-search (visual processing alone). The same idea of leading level can help the understanding of other aspects of our data. Thus, whereas intention to memorize does not play a dominant role in the present studies, there is a surprising degree of conscious access to the processes involved in many of the conditions. For example, perceptual processing has often been considered to be highly automatic and “cognitively impenetrable” (Pylyshyn, 1984), but it seems that this level of processing can become quite accessible if a task conveys the leading role to it, as in Experiments 2 and 3.

In general, it may be argued that retrieval tasks that rely on relatively deep forms of processing, and that are only loosely focused with respect to their leading level,

will benefit from all previously encoded information (or previously activated processes) at and above the lowest level that they can utilize. Such retrieval tasks will demonstrate the graded pattern of LOP function (1D in Table 1). Tasks that draw only on the currently shallowest form of processing will benefit only from previous activation of these lowest processes, and such tasks will therefore demonstrate the flat LOP pattern across encoding conditions. Finally, retrieval tasks that are narrowly focused on their leading level will not profit from lower levels of activation, will benefit from their specifically required type of information, but will not derive further benefit from types of information that are located at higher levels of the processing hierarchy. These retrieval tasks will therefore demonstrate the stepped LOP pattern that combines features of strictly modular and hierarchical models.

Although we are arguing that simple dichotomies such as perceptual/conceptual and direct/indirect are insufficient to account for the data adequately, nonetheless these ideas may provide a useful first approximation to a description of the findings. For example, the direct/direct distinction highlights two broad categories of goals in priming and memory retrieval tasks. The first set of goals is found in indirect tasks; the goal here is simply to comply with task instructions—to generate a rhyming word, complete a word fragment, etc. The second goal is found in direct tasks, and this is to retrieve (or generate) a specific item—namely, an item on the original presentation list. These notions lead to the generation/recognition model that explains dissociations between direct and indirect word fragment cued tests (e.g., Challis & Tulving, 1993; Jacoby & Hollingshead, 1990). Both direct and indirect tests involve some generation from the presented cue word, and this generation phase is facilitated by sensory similarity between cue and target (Craik et al., 1994; Roediger et al., 1992; Weldon, Roediger, & Challis, 1989). While any generated response is “correct” in an indirect test, a direct test requires that the generated candidates be subjected to a further recognition phase. In this case all types of encoded information that affect recognition will benefit performance. In general then, indirect tests are sensitive to lower-level types of information and show flat LOP functions, whereas direct tests are also sensitive to higher-level processes and show graded LOP effects. One exception here involves the direct tasks in Experiment 3 which show stepped functions; in this case we assume that the cues do *not* lead to a generation process, but that the salient aspect of the cue (i.e., its graphemic, phonological, or semantic properties, depending on the task in question) is used to locate a list item that shares that property.

What aspects of the present data encourage the notion of a hierarchy or perhaps heterarchy (see Turvey, Shaw, & Mace, 1978) of processing levels rather than a set of semiautonomous processing modules? In our view the critical evidence lies in the fact that out of 13 retrieval task conditions reported in the three experiments, all but 1 show functions that are flat, stepped, or graded. On the assumption that Intentional encoding lies between Living and Self encoding in Experiment 1, only the Free Recall condition yields a function that declines from a lower to an ostensibly higher level of encoding, and this one exception can be explained on the basis of intentional encoding giving rise to a greater degree of organization which benefits free recall. If the underlying cognitive architecture was strictly modular, and retrieval tasks were tuned to respond optimally to one type of information, it would be expected that performance would fall off following both lower levels and encoded higher levels

of encoding. The present data show no such effect. Rather it seems that, as a rule, lower perceptual levels are also invoked and activated when higher-level processes are induced. The heterarchical nature of the cognitive architecture is manifested in its exceptional flexibility; in response to task demands it ranges broadly from the weakly parallel (1B in Table 1) to the strongly hierarchical (1D in Table 1) mode of operation.

Another emerging independent source of evidence for the type of cognitive architecture we are dealing with is neurophysiology, in particular brain imaging data. It has been shown that there are brain structures demonstrating different rates of growth in phylogenesis and that these structures build anatomical gradients in their cortical expansion from posterior to anterior to prefrontal regions (Deacon, 1996; Goldberg, 1989). An interesting question for future research is whether the same gradients of activation can be shown in behavioral experiments (see, for example, Kapur, Craik, Tulving, Wilson, Houle, & Brown, 1994; Velichkovsky, 1994). If the answer is positive, then LOP effects at the behavioral level may be rephrased in terms of developmental and even evolutionary principles. Such a "new look" version of levels of processing may therefore lead to a reversal of theoretical perspective from information processing depths to evolutionary heights and to an integration of behavioral findings with theory and data from neurophysiological and developmental studies.

ACKNOWLEDGMENTS

The research reported in this article was supported in part by grants from the German Research Foundation (DFG SFB360/B4) to Boris M. Velichkovsky and from the Natural Sciences and Engineering Research Council of Canada to Fergus I. M. Craik and Endel Tulving. The authors are grateful to Wendy Armstrong, Esther Meisels, and Ruth Sirur for technical help.

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Received November 2, 1995